

System Architecture Diagram

DHT11 -> Arduino Uno -> LCD / USB Serial -> PC publisher -> MQTT broker -> web dashboard

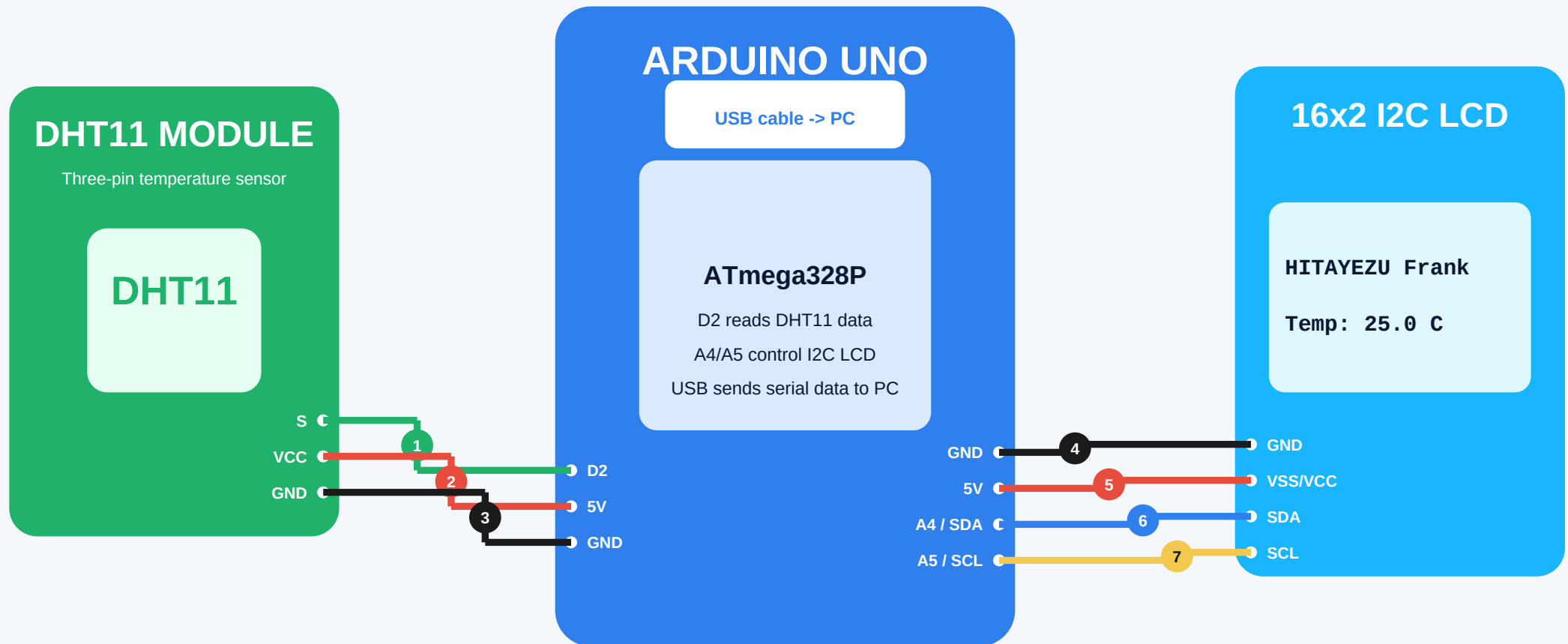


MQTT topic [rca/hitayezu-frank-duff/temperature](#)

Retained JSON contains device, sensor, student, temperature, unit and UTC timestamp.

Seven-Wire Hardware Connection

DHT11 uses three wires; the I2C LCD backpack uses four wires



Wire summary

- | | | | |
|-------------------------|-----------------------|------------------------|----------------------|
| 1 DHT11 S -> UNO D2 | 2 DHT11 VCC -> UNO 5V | 3 DHT11 GND -> UNO GND | 4 LCD GND -> UNO GND |
| 5 LCD VSS/VCC -> UNO 5V | 6 LCD SDA -> UNO A4 | 7 LCD SCL -> UNO A5 | |

Safety: disconnect USB while wiring. Verify that the LCD pin called VSS/VCC by the kit is really its +5 V supply pin.